

Data

Simon Singh, The Black Chamber

A comprehensive guide to encryption with interactive examples for beginners.

Unicode Consortium

Information on world scripts and emojis.

Lingojam Binary Translator

Translate English text into the binary representation used by a computer.

The Logic Lab

Explore logic gates by creating interactive circuits.

Studio.code.org: Binary Game

See how binary code works by playing this fun game.

OpenLearn: Analogue Universe, Digital Worlds

A free online course on the differences between analogue and digital, and how analogue information is represented by a computer in the form of numbers.

Khan Academy: Cryptography

Short YouTube lectures and reading material on cryptography, from ancient times to the modern day.

Programming techniques

Codecademy

Online interactive tutorials to learn programming languages such as Python, Java, HTML, and CSS.

Computerphile YouTube Channel

Beginner-friendly videos where experts explore the inner workings of computers. Includes computer history.

HackerRank

Fun coding challenges and global coding competitions.

Stack Overflow

The world's largest developer community, with extensive forums to answer questions and help troubleshoot problems.

TutorialsPoint

Simple, clear explanations on a variety of technical subjects.

Tynker

Online platform that introduces kids to programming.

Programming languages

CoderDojo

A global network of free, volunteer-led, community-based programming clubs for young people.

EarSketch

A free, educational programming environment that seeks to teach people how to code in Python and JavaScript through music composing and remixing.

W3 Schools

Tutorials on a wide range of programming languages used to create websites.

Code Club International

An international network of volunteers and educators who run free coding clubs for young people aged 9–13.

Code.org

Learn to code with fun tutorials or try out coding in an "Hour of Code".

Scratch

A drag-and-drop programming language ideal for kids and beginners, with a global community that shares projects.

Python.org

The online hub for the open-source programming language Python, with guides, third-party applications, and general information.

Cplusplus.com

Information, tutorials, and articles on the C++ programming language.

Javascript.com

Resources, news, and guides on the JavaScript programming language.

Kodu Game Lab

Create, share, and learn about the Kodu programming language.

Ruby

Downloads, documentation, libraries, and general information about the Ruby programming language.

Stack Overflow

Developers give advice to other programmers on coding issues.